# TANMAY DESHPANDE

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## SUMMARY

Software engineer working across backend services and AI/LLM tooling, with production experience in observability and test automation at UBS and peer-reviewed research in applied optimisation.

## SKILLS

**Languages:** Python, Go, TypeScript, SQL, Bash, Kotlin
**AI & LLM:** LangGraph, RAG (ChromaDB, HuggingFace embeddings), Transformers.js & Whisper, LLM evaluation harnesses, OpenRouter / Ollama APIs
**Backend & Infra:** FastAPI, Express, Chi, PostgreSQL, SQLite, Prisma, sqlc, Docker / Compose, MinIO (S3), OpenTelemetry, Prometheus, Grafana, Playwright
**Frontend:** React, Vite, Tailwind CSS

## EXPERIENCE

**UBS** — Hyderabad, India
*Software Engineer* — *Aug 2026 – Present*

- Built end-to-end test coverage for an internal sensitive-data document-scanning platform using **Playwright** across **15 critical flows**, wired into the team's CI pipeline.
- Building automated VM provisioning for a newly formed team (MVP): integrating an internal server-lifecycle API to create/delete virtual machines and trigger Amelia automation jobs, alongside supporting databases and test pipelines.

**UBS** — Hyderabad, India
*Software Engineering Intern* — *Jan – Jul 2026*

- Instrumented the document-scanning service (a platform handling ~9,700 scans/month) using an OpenTelemetry-based internal library, surfacing metrics on a **Prometheus + Grafana** dashboard.
- Built a remediation-notification service (internal Microsoft Teams bot wrapper + email API) that queries remediated findings and each user's last-notified state from the database and routes alerts to the correct owner.

## PUBLICATION

Menon, B. K., **Deshpande, T.**, Pal, A., & Kothandaraman, S. (2025). "Critical regions identification and coverage using optimal drone flight path planning for precision agriculture." *Results in Engineering* (Elsevier), 104081. doi:10.1016/j.rineng.2025.104081 — **16 citations**

- Co-authored a peer-reviewed pipeline combining image pre-processing, **K-means** clustering and **2-opt** TSP heuristics for drone coverage routing, reporting a **22–46%** runtime improvement over baselines.

## PROJECTS

**TaskTide** — Distributed workflow platform
*Go, PostgreSQL, MinIO/S3, React, Docker*

- DAG workflows designed in a drag-and-drop editor and executed across a self-registering worker fleet, with dependency-aware scheduling via topological sort and unblocked-task queries.
- Persists workflow/task state in Postgres (sqlc + goose migrations), passes task data as files through MinIO/S3, and deploys with a single Docker Compose command.

**brickbench** — LLM debloat-safety evaluation harness
*Python, SQLite, ADB / Android emulator*

- Applies LLM-proposed Android debloat plans to a booted emulator snapshot and verifies them with functional probes; results are **SHA-256** hashed for reproducibility.
- Uses a semaphore-gated emulator pool with atomic run claiming so parallel workers never double-execute, and feeds verified verdicts into a learned safe-list.

**ai-workflow-platform** — Governed AI workflow platform
*FastAPI, LangGraph, ChromaDB, SQLAlchemy*

- LLM-assisted workflow drafting over an uploaded regulatory knowledge base (RAG), with deterministic rules making the final decision and a replayable audit trail.
- JWT + RBAC authentication, immutable versioned workflows, and account-level run serialisation.

## EDUCATION

**Vellore Institute of Technology** — Chennai, India
*B.Tech, Computer Science & Engineering (AI & ML)* — *2022 – 2026*
GPA: 8.49/10

## ADDITIONAL

Microsoft Certified: Azure AI Fundamentals (**AI-900**) · Languages: English, Hindi, Marathi